

MICROWAVE AND RF DESIGN SOFTWARE

Some of the most commonly used software to design accurate, high-performance microwave and RF circuits

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Electronic gadgets like cellphones, FM receivers, GPS receivers and radar warning equipment work on radio-frequency (RF) or microwave signals. Designing these gadgets requires a high precision. Thus a need arises to take help of software.

Says Pieter L.D. Abrie, managing director of Ampsa, "Electronic components are modelled in software and the performance that can be expected by combining different components can be predicted."

Shawn Carpenter, vice president, sales and marketing, Sonnet Software, says, "Microwave and RF software tools predict very accurately the degree of power transfer, circuit mismatch, electromagnetic radiation and electrical resonances of these circuits."

Manoj Menghrajani, product manager-FEKO, ni2designs, adds, "The advantage of using simulation software for microstrip and waveguide devices is that engineers can test and optimise designs before building prototypes, increasing their confidence that prototypes will perform as expected."

Detailed below are some of the software that can be used for microwave and RF designing.



FEKO is an electromagnetic analysis software suite that helps the users to solve a wide variety of electromagnetic problems. The latest version of the software is FEKO Suite 5.5.

What's special? FEKO software combines a wide range of mathematical methods for simulation of micro-

wave and EM problems. The generally applicable and powerful multilevel fast multipole method and hybridisation of the methods of moments (MoM) with FEM and asymptotic high-frequency physical optics, geometrical optics and uniform theory of diffraction techniques makes FEKO particularly suitable for solving electrically large problems and thus a better proposition than traditional codes.

Also, hybridisation implies that these solution techniques can be applied to different parts of the same model to optimise the solution time and results, significantly improving the behaviour of the memory and run-time requirements, resulting in enormous computational savings.

FEKO simulations run in parallel on all the cores of a multi-core CPU, without incurring extra licence costs.

Home Page: <http://www.feko.info/>
Download URL: http://www.feko.info/Download_Centre
Platform Supported: Windows, Linux

Sonnet Lite



Sonnet Lite is an EM-analysis tool. It is a free, feature-limited version of Sonnet's professional Sonnet Suite. The latest version of Sonnet Lite is 12.53.

What's special? Sonnet Lite allows RF and microwave designers to incorporate in a circuit up to three ideal surface-mount devices (SMDs). It is useful for simulation of (ideal) surface-mount inductors, capacitors and resistors. The effects of the SMD terminal contact width on the circuit, as well as interaction with the surface-mount pad may be included, and this sometimes has a significant electrical effect at higher frequencies.

Sonnet Lite also includes seamless interfaces to popular RF electronic design automation (EDA) frameworks from Agilent (Agilent ADS) and Applied Wave Research (AWR Microwave Office). You can use it within these frameworks for modelling microwave circuits, or use Sonnet's internal project editor for drawing circuits, vias, filters or other 3-D planar circuit layouts.

Home Page: <http://www.sonnetsoftware.com/>
Download URL: <http://www.sonnetsoftware.com/products/lite/download.html>
Platform Supported: Windows

Multimatch IMW

Multimatch IMW is an impedance-matching software that can be used to synthesise lumped-element, distributed or mixed lumped/distributed matching networks. The latest version of the software available is 9.1.

What's special? Multimatch IMW synthesises high-quality RF and microwave impedance-matching networks up to microstrip level. While working with the software, microstrip lines are assumed to be lossless during synthesis. Pads and connecting lines, as well as parasitic inductance for capacitors and parasitic capacitance for inductors, can be specified for the networks to be synthesised. The accuracy of the software can be extended up to multimetre-wave frequencies by the user.

Home Page: <http://www.amps.com/>
Download URL: <http://www.amps.com/fp1MW.htm>
Platform Supported: Windows ●

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